

## AP CALCULUS AB – UNIT 5

5.2–5.5 | Derivative Graph Analysis  
- Part 1

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

## Lesson Overview and Learning Objectives

- Find critical numbers and intervals of increase and decrease.
- Use the First Derivative Test to classify local extrema.
- Apply the Extreme Value Theorem and closed-interval method for absolute extrema.
- Connect the sign of  $f'$  with the shape of  $f$ .

## Keys: Essential Knowledge and Vocabulary

## Critical number

An input  $c$  in the domain of  $f$  where  $f'(c) = 0$  or  $f'(c)$  does not exist.

## Increasing/decreasing

$f' > 0$  gives increasing;  $f' < 0$  gives decreasing on an interval.

## First Derivative Test

$+\rightarrow-$  gives a local maximum;  $-\rightarrow+$  gives a local minimum.

## Extreme Value Theorem

A continuous function on a closed interval attains absolute maximum and minimum values.

## Closed-interval method

Evaluate  $f$  at endpoints and all interior critical numbers; compare the values.

## Candidate distinction

A critical number is only a candidate; classification requires sign change or another valid test.

## Explanation, Strategy, and Common Errors

- Find the domain first, then solve  $f' = 0$  and include points in the domain where  $f'$  is undefined.
- Build a sign chart for  $f'$  using test inputs or factor signs.
- State extrema as points or values as requested, and distinguish local from absolute.
- For an absolute-extrema problem on  $[a, b]$ , endpoints are mandatory candidates even though derivatives are not required there.
- When given the graph of  $f'$ , local extrema of  $f$  occur at sign changes, not merely at zeros of  $f'$ .

## Solved Example: First derivative sign chart

For  $f(x) = x^3 - 3x^2 - 9x + 1$ , find intervals of increase/decrease and local extrema.

## Answer and Reasoning

$f' = 3(x - 3)(x + 1)$ . Increasing on  $(-\infty, -1) \cup (3, \infty)$ , decreasing on  $(-1, 3)$ . Local max at  $x = -1$  with  $f(-1) = 6$ ; local min at  $x = 3$  with  $f(3) = -26$ .

## Solved Example: Critical derivative undefined

Analyze  $f(x) = x^{2/3}$  near 0.

**Answer and Reasoning**

$f' = \frac{2}{3}x^{-1/3}$  is undefined at 0, which is in the domain. The derivative changes from negative to positive, so  $f$  has a local and absolute minimum at  $(0, 0)$ .

**Solved Example: Closed interval**

Find absolute extrema of  $f(x) = x^3 - 3x$  on  $[-2, 2]$ .

**Answer and Reasoning**

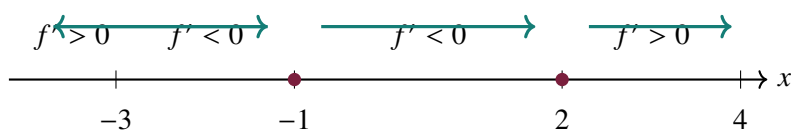
Critical numbers  $x = \pm 1$ . Values:  $f(-2) = -2$ ,  $f(-1) = 2$ ,  $f(1) = -2$ ,  $f(2) = 2$ . Absolute max 2 at  $x = -1, 2$ ; absolute min  $-2$  at  $x = -2, 1$ .

**Solved Example: Graph of  $f'$** 

Using the shared sign chart, classify the critical numbers.

**Answer and Reasoning**

At  $x = -1$ ,  $f'$  changes  $+$   $\rightarrow$   $-$ , so  $f$  has a local maximum. At  $x = 2$ ,  $f'$  changes  $-$   $\rightarrow$   $+$ , so  $f$  has a local minimum.

**Shared Representation / Data****AP Exam Language**

For theorem applications, state every hypothesis. For graph analysis, justify sign changes and compare all required candidates. For optimization, state the objective, constraint, feasible domain, and verification.

**Leveled Practice****Easy Foundations**

Build fluency with the definition, notation, and one-step applications.

1. Find critical numbers of  $f(x) = x^2 - 6x + 1$ .

2. If  $f' > 0$  on  $(1, 5)$ , describe  $f$  there.

3. What sign change of  $f'$  gives a local minimum?

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4. List candidates for absolute extrema on  $[a, b]$ .

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### Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

5. For  $f' = x(x - 4)$ , determine intervals where  $f$  increases.

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6. Classify  $x = 0$  and  $x = 4$  from the previous derivative.

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7. Find absolute extrema of  $f(x) = x^2 - 4x + 5$  on  $[0, 5]$ .

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8. If  $f'(c) = 0$  but  $f'$  is positive on both sides, classify  $c$ .

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### Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

9. For  $f(x) = x + 4/x$  on  $(0, \infty)$ , find the absolute minimum.

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10. For  $f'(x) = (x + 2)^2(x - 1)$ , classify critical numbers.

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11. A continuous function has one critical number in  $(a, b)$ . Is its absolute maximum guaranteed to occur there?

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12. If  $f$  is increasing on  $(-\infty, 2)$  and decreasing on  $(2, \infty)$ , what can be said about  $x = 2$  if  $f$  is defined there?

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### Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

13. Find absolute extrema of  $f(x) = x^{2/3}(x - 4)$  on  $[-1, 8]$ .

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14. Suppose  $f'$  is zero on an entire interval  $[1, 3]$ . Describe  $f$  there and classify its points.

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15. If  $f'$  changes  $+$   $\rightarrow$   $-$  at  $c$  but  $f(c)$  is undefined, can  $f$  have a local maximum at  $c$ ?

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**Challenge / AP Extension**

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

16. Construct a differentiable function with a critical point at 0 that is neither maximum nor minimum.

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17. Prove that a differentiable function with  $f' > 0$  everywhere on an interval has at most one zero there.

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## AP-Style Multiple-Choice Practice

## Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. A critical number must

- (A) make  $f' = 0$  only
- (B) be in the domain of  $f$  and make  $f' = 0$  or undefined
- (C) be an endpoint
- (D) be a local extremum

2. If  $f'$  changes from negative to positive at  $c$ , then  $f$  has

- (A) a local maximum
- (B) a local minimum
- (C) an inflection point necessarily
- (D) no conclusion

3. For absolute extrema on  $[a, b]$ , one must compare

- (A) critical numbers only
- (B) endpoints only
- (C) endpoints and interior critical numbers
- (D) zeros of  $f$

4. If  $f'(c) = 0$ , then  $c$

- (A) must be a maximum
- (B) must be a minimum
- (C) is a critical number but may be neither extremum
- (D) is not in the domain

## Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. For  $f(x) = x^3 - 6x^2 + 9x$  on  $[0, 5]$ , the absolute maximum occurs at

- (A)  $x = 0$
- (B)  $x = 1$
- (C)  $x = 3$
- (D)  $x = 5$

6. If  $f'(x) = (x - 1)(x + 3)$ ,  $f$  decreases on

- (A)  $(-\infty, -3)$
- (B)  $(-3, 1)$
- (C)  $(1, \infty)$
- (D) all real numbers

## AP-Style Free-Response Practice

### No Calculator

Let  $f(x) = x^4 - 4x^3$  on  $[-1, 5]$ .

(a) Find the critical numbers. [2 points]

(b) Determine intervals of increase and decrease and classify local extrema. [3 points]

(c) Find absolute extrema on  $[-1, 5]$ . [3 points]

### Calculator Permitted

The graph of  $f'$  is sampled at selected points:  $f'(-2) = 3$ ,  $f'(0) = -1$ ,  $f'(2) = -4$ ,  $f'(4) = 2$ , with zeros at  $x = -1$  and  $x = 3$ .

(a) Identify intervals on which  $f$  increases and decreases. [2 points]

(b) Classify the critical points. [2 points]

(c) If  $f(-1) = 7$  and  $f(3) = 1$ , can the absolute maximum on  $[-2, 4]$  be identified without endpoint values? [2 points]